
Leant

a Djex-based synthesis REPL for Lean 4

an overview and tutorial, with a detailed tour of `:synth`

An **experimental** tool in active development. Interfaces, output, and capabilities change frequently.

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1 Introduction

Leant is an interactive read–eval–print loop for Lean 4, in the spirit of GHCi: you type expressions and they are evaluated, you type declarations and they enter the session, and a family of colon-commands gives you type queries, documentation, search, interactive proving, and automatic *term synthesis* with machine-checked answers. The synthesis command takes up half of this document.

Lean 4 itself is normally driven from an editor: the language server shows goals and diagnostics as you edit a file. Leant complements that workflow with a conversational one. It is aimed at exploration (trying a lemma, poking at a definition, asking *is this type even inhabited?*), where the unit of work is a line, not a file.

experimental

Leant is an **experimental project in active development**. It is a research vehicle, not a supported product: commands appear, change shape, and disappear between commits; output formats are not stable; and some features are bounded approximations by design. Every transcript in this document was produced by the tool at the date on the title page and may not match the tool you build tomorrow. When this document and `:help` disagree, believe `:help`.

Leant is implemented in Haskell and speaks the backend protocol directly; term synthesis (§5) embeds the Djex synthesis library in-process.

Under the hood, Leant drives the community `leanprover-community/repl` binary over a JSON protocol: every input becomes a command against a persistent Lean environment, and every claim Leant makes, including every synthesized term, is checked by that backend before it is shown. Sessions survive backend crashes (the session history replays automatically), can be saved and restored (`:pickle/:unpickle`, including synthesis-ready companions for Leant-created snapshots), and can be recorded as transcripts.

2 Getting started

Build the binary once with `cabal build exe:leant` (GHC 9.12.4; the vendored `lib/Djex` submodule must be initialized), then run `leant.cmd`, or the built executable directly. Leant finds the Lean backend the same way `LeanInteract`’s cache laid it out, or accepts an explicit `--repl-exe/LEANT_BACKEND`. Without a Lake project it runs against the plain Lean 4 standard library, which is also the least memorable thing about it: no imports, subsecond startup.

A session begins as a calculator and escalates quietly:

session

```
λ> 2 + 2
4
λ> it * 10
40
λ> def double (n : Nat) : Nat := n + n
λ> double 21
42
```

```
λ> :type double
double : Nat → Nat
```

Expressions are tried with `#eval` and fall back to `#check`; the last evaluated value is bound as `it`, GHCi-style. Declarations (`def`, `theorem`, `inductive`, ...) extend the session environment, and any `#`-command (`#eval`, `#check`, `#print axioms`) passes straight through. Multi-line input opens automatically when a line is syntactically incomplete (type a structure header and just keep going), and an empty line submits it.

3 The commands

3.1 Session and evaluation

<code>:help, :h, :?</code>	show the command summary
<code>:quit, :q</code>	exit
<code>:type EXPR, :t</code>	show the type of EXPR (<code>#check</code>)
<code>:info NAME, :i</code>	show the definition of NAME (<code>#print</code>)
<code>:load FILE, :l</code>	reset the session and load a <code>.lean</code> file
<code>:reload, :r</code>	reload the last loaded file
<code>:import MOD</code>	add an import (rebuilds the session; unavailable over an opaque snapshot)
<code>:imports</code>	list active imports, or imports restored by <code>:reset</code>
<code>:undo</code>	revert the last state-changing command
<code>:reset</code>	clear definitions or a snapshot base (keeps configured imports)
<code>:history</code>	show commands after the current import/snapshot base
<code>:set OPT VAL</code>	a Lean <code>set_option</code> that persists in the session

3.2 Discovery

<code>:browse [NS]</code>	list declarations in a namespace, or the session's own
<code>:browse! NS</code>	...including compiler-generated auxiliaries
<code>:doc NAME</code>	show a docstring
<code>:search TEXT</code>	search declaration names, case-insensitively
<code>:search? TYPE</code>	proof search: what proves TYPE? (via <code>exact?</code>)

3.3 Proving and synthesis

<code>:prove [PROP]</code>	prove PROP interactively, tactic by tactic (without an argument: resume the last sorry)
<code>:synth TYPE</code>	synthesize verified terms of TYPE; candidates are bound as <code>it1</code> (= <code>it</code>), <code>it2</code> , ... (§5)
<code>:set synth-engine E</code>	<code>djinn</code> (default) <code>exference</code> <code>both</code>
<code>:set synth-steps N</code>	Exference's step budget (default 4096)
<code>:set synth-classical B</code>	offer classical candidates for constructively refuted goals (on/off, default on)

```
:set synth-library B    library premises for recursive inductives (on|off, default on; §5.6)
```

3.4 Persistence and utility

```
:transcript [F|on|off] record a transcript of the session
:timestamps [on|off]    timestamp transcript entries
:pickle FILE            save the environment and synthesis companion
:unpickle FILE          restore an environment as a new undo/history base
:env                    show the backend environment id
:time                   toggle per-command timing
:!  
:! CMD                  run a shell command
```

4 Interactive proving

`:prove` turns the prompt into a tactic loop against the backend’s proof-state protocol. Every line is a tactic; `:undo` pops one step; `:qed NAME` assembles the accumulated script into a theorem and replays it into the session, so the result is usable afterwards. Each displayed proof state is followed by an automatically discovered tactic suggestion that Lean has already checked. The candidate probes are shaped by the goal and its hypotheses: an intro suggestion names the binders it would introduce, a disjunction goal is probed with `left/right`, a hypothesis whose type a single step can take apart is probed with `cases` or `obtain`, and a data-typed variable the goal mentions is probed with `induction`. The search prefers a candidate that closes the goal outright and annotates the suggestion accordingly (`closes the goal`, `splits into 2 goals`); when no single tactic closes it, a second phase chains quick finishers onto the best progressing candidates (`constructor <;> omega`, or an `obtain` step chained with the `exact` it enables), so even the opening suggestion is often a complete checked proof. The chains also try `simp_all`, unfolding the definitions the goal mentions and calling in `omega` on whatever arithmetic remains, which is how an induction suggestion can arrive as a finished proof: `induction l <;> simp_all [myLen]`, or `induction n <;> simp_all [double] <;> omega`. The suggestion is advisory—it does not advance the state or enter the script—and `:suggest` reprints it without rerunning the search. Below, the very first suggestion is already a complete proof; the session ignores it and proceeds step by step instead. Proof-state identifiers are local to one backend process; if it stops, Leant leaves prove mode and prints the accumulated script rather than submitting a stale identifier after session reconstruction.

session

```
λ> :prove ∀ p q : Prop, (p → q) → p → q ∧ p
entering prove mode – type tactics; :help for commands
⊢ ∀ (p q : Prop), (p → q) → p → q ∧ p
suggestion: exact fun p q a a_1 => (a a_1, a_1) (closes the goal)
⊢> intro p q h hp
p q : Prop
h : p → q
hp : p
⊢ q ∧ p
```

```
suggestion: exact (h hp, hp) (closes the goal)
├> exact (h hp, hp)
All goals accomplished
finish with :qed [NAME], inspect with :script
├> :qed mp_and
saved: theorem mp_and : ∀ p q : Prop, (p → q) → p → q ∧ p
```

Accepting a suggestion is just typing it. On a conjunction of two arithmetic facts, the opening suggestion is already the whole chained proof, and the session is two lines long:

session

```
λ> :prove (∀ a b : Nat, a + b = b + a) ∧ (∀ a b : Nat, a + 2 * b = b + b + a)
entering prove mode – type tactics; :help for commands
├ (∀ (a b : Nat), a + b = b + a) ∧ ∀ (a b : Nat), a + 2 * b = b + b + a
suggestion: constructor <;> omega (closes the goal)
├> constructor <;> omega
All goals accomplished
finish with :qed [NAME], inspect with :script
├> :qed two_omegas
saved: theorem two_omegas : (∀ a b : Nat, a + b = b + a) ∧ (∀ a b : Nat, a + 2 * b = b
↪ + b + a)
```

That chain was suggested because *both* halves fall to omega. When they need different tools, the suggestion adapts to whichever goal is in focus — arithmetic for the first half, a found library lemma for the second:

session

```
λ> :prove (∀ a b : Nat, a + b = b + a) ∧ (∀ a b : Nat, a * b = b * a)
entering prove mode – type tactics; :help for commands
├ (∀ (a b : Nat), a + b = b + a) ∧ ∀ (a b : Nat), a * b = b * a
suggestion: constructor (splits into 2 goals)
├> constructor
– goal 1 of 2 –
case left
├ ∀ (a b : Nat), a + b = b + a
– goal 2 of 2 –
case right
├ ∀ (a b : Nat), a * b = b * a
suggestion: omega (closes the goal)
├2> omega
case right
├ ∀ (a b : Nat), a * b = b * a
suggestion: exact fun a b => Nat.mul_comm a b (closes the goal)
├> exact fun a b => Nat.mul_comm a b
All goals accomplished
finish with :qed [NAME], inspect with :script
├> :qed add_mul_comm
saved: theorem add_mul_comm : (∀ a b : Nat, a + b = b + a) ∧ (∀ a b : Nat, a * b = b *
↪ a)
```

The probes are shaped by the hypotheses as much as by the goal. An existential hypothesis draws an obtain suggestion that names its witness; once the conjunction inside is the thing in the way, the suggested step is the obtain that splits it, chained with the exact it enables:

session

```
λ> :prove (∃ x : Nat, 2 < x ∧ x < 4) → ∃ y : Nat, 2 < y
entering prove mode – type tactics; :help for commands
⊢ (∃ x, 2 < x ∧ x < 4) → ∃ y, 2 < y
suggestion: intro h
⊢> intro h
h : ∃ x, 2 < x ∧ x < 4
⊢ ∃ y, 2 < y
suggestion: obtain (x, h1) := h
⊢> obtain (x, h1) := h
x : Nat
h1 : 2 < x ∧ x < 4
⊢ ∃ y, 2 < y
suggestion: obtain (h, h2) := h1 <;> exact Exists.intro x h (closes the goal)
⊢> obtain (h, h2) := h1 <;> exact Exists.intro x h
All goals accomplished
finish with :qed [NAME], inspect with :script
⊢> :qed ex_two_lt
saved: theorem ex_two_lt : (∃ x : Nat, 2 < x ∧ x < 4) → ∃ y : Nat, 2 < y
```

A suggestion is not limited to one step. Here the goal needs structural induction on a user-defined function; once intro brings the variable into the context, the suggested chain arrives as a complete verified proof:

session

```
λ> def myLen : List Nat → Nat
...> | [] => 0
...> | _ :: t => myLen t + 1
...>
λ> :prove ∀ l : List Nat, myLen l = l.length
entering prove mode – type tactics; :help for commands
⊢ ∀ (l : List Nat), myLen l = l.length
suggestion: intro l
⊢> intro l
l : List Nat
⊢ myLen l = l.length
suggestion: induction l <;> simp_all [myLen] (closes the goal)
⊢> induction l <;> simp_all [myLen]
All goals accomplished
finish with :qed [NAME], inspect with :script
⊢> :qed myLen_length
saved: theorem myLen_length : ∀ l : List Nat, myLen l = l.length
```

The same pattern scales to arithmetic recursion. The chain below was assembled from three of the probes at once: induction because the goal mentions a data-typed variable, simp_all unfolding the definition the goal is about, and omega for the arithmetic the simplifier leaves behind – a finished proof of a small theorem about a function defined moments earlier:

session

```

λ> def double : Nat → Nat
...> | 0 => 0
...> | n + 1 => double n + 2
...>
λ> :prove ∀ n : Nat, double n = 2 * n
entering prove mode – type tactics; :help for commands
⊢ ∀ (n : Nat), double n = 2 * n
suggestion: intro n
⊢> intro n
n : Nat
⊢ double n = 2 * n
suggestion: induction n <;> simp_all [double] <;> omega (closes the goal)
⊢> induction n <;> simp_all [double] <;> omega
All goals accomplished
finish with :qed [NAME], inspect with :script
⊢> :qed double_two_mul
saved: theorem double_two_mul : ∀ n : Nat, double n = 2 * n

```

A sorry in an ordinary declaration prints its goal and offers the same loop via bare `:prove`. Prove mode also cooperates with `:synth`; §5.9 shows what that buys.

5 Term synthesis: `:synth`

`:synth TYPE` answers the question “*write me a term of this type*”; read through propositions-as-types, that is “*prove this*.” Sometimes it answers the stronger question “*show me that no such term exists*.” It is built on the vendored Djex library, linked in-process. Djex began by merging two classic Haskell program synthesizers behind one checked synthesis foundation: **Djinn**, a complete and terminating proof search for intuitionistic propositional logic (Dyckhoff’s LJ_T calculus) that emits programs, and **Exference**, a ranked heuristic search under explicit budgets. It has since moved well beyond both originals. It fixes a long list of their bugs; the worst of them let a search that had merely approximated a type report that type uninhabited, and an exhausted search over an approximated space now counts for nothing. It implements much better type-class support, with a unified, validated constraint contract shared by both engines: Exference resolves givens, superclasses, and instances, while Djinn checks contexts and synthesizes dictionary-independent terms. And it adds a growing level of support for rank-*N* and impredicative types: goal-side quantifiers open through polarity-aware plan families, quantified hypotheses are instantiated at a bounded set of sequent-supplied types, and impredicative instantiation is admitted under a guard that never invents a polytype the query did not supply. Djinn’s current context-free hypothesis rule reaches four leading binders without raising its fixed search caps; chains with five or more stay opaque. Its singleton, pairwise, and triple open/opaque frontiers grow cubically and cover every choice through seven independent quantified sites without enumerating a power set. That scope is still being improved, and Leant keeps the boundary to the engines narrow on purpose, so each improvement arrives by version bump. The engines speak Haskell types; Leant translates Lean goals into their fragment and their answers back into Lean.

Three rules run through the design:

- **The engine is never trusted.** Every candidate is re-elaborated by the Lean backend against the exact goal (example : (T) := term) before you see it. A rendering bug or an engine bug costs a dropped candidate, never a wrong answer.
- **Refusals come with reasons.** A goal outside the supported fragment is turned away with a note saying what fell outside, not quietly mangled into something answerable.
- **Negative verdicts are labeled by strength.** “Provably uninhabited” appears only when the translation was complete and lossless; anything weaker is reported as “no term found within bounds,” which claims nothing.

Transcripts in this section are lightly abridged: trailing candidates beyond the ones shown, and trailing truncation notes, are elided. Every line that does appear is a verbatim record of a live session.

5.1 Higher-order plumbing

The sweet spot is the structural fragment: arrows, products, sums, $\perp$, $\top$, negation, *Iff*, and quantifiers over opaque variables. These are the “plumbing” terms one writes constantly. Free capital identifiers are auto-bound, so quick queries stay quick:

session

```
λ> :synth ((A → B → C) → (A → B) → A → C)
  it1 fun f g x => f x (g x)
λ> :synth (A × (B ⊕ C) → (A × B) ⊕ (A × C))
  it1 fun ⟨x, y⟩ => match y with | .inl z => .inl ⟨x, z⟩ | .inr w => .inr ⟨x, w⟩
λ> :synth (∀ p q : Prop, (p ↔ q) → ¬p → ¬q)
  it1 fun _ _ (⟨_, f⟩ k x => k (f x))
```

The first is the *S* combinator; the second distributes a product over a sum; the third is contraposition across an *Iff*, with the backward implication projected out by the pattern $\langle _, f \rangle$. Binders are named by role (functions *f g h*, values *x y z*, negations and continuations *k*), a small thing that makes large candidates much easier to read.

The everyday combinators all come out on the first try, and each first candidate is the term you would have written yourself — *flip*, *composition*, *curry*, *uncurry*, and *product associativity*:

session

```
λ> :synth ((a → b → c) → b → a → c)
  it1 fun f x y => f y x
λ> :synth ((b → c) → (a → b) → a → c)
  it1 fun f g x => f (g x)
λ> :synth ((A × B → C) → A → B → C)
  it1 fun f x y => f ⟨x, y⟩
λ> :synth ((A → B → C) → A × B → C)
  it1 fun f ⟨x, y⟩ => f x y
λ> :synth (((A × B) × C) → A × (B × C))
  it1 fun ⟨⟨x, y⟩, z⟩ => ⟨x, ⟨y, z⟩⟩
```

Candidates are ranked smallest-first and *bound into the session* as `it1`, `it2`, ..., with bare `it` the best one. They are ordinary definitions; you can evaluate them immediately:

```
session
λ> :synth (a → a → a)
  it1 fun _ x => x
  it2 fun x _ => x
λ> #eval it2 "left" "right"
"left"
```

Both inhabitants of $a \rightarrow a \rightarrow a$ that matter, and picking one is just using its name. For programs, as opposed to proof-irrelevant propositions, the choice between candidates is the whole point.

An Iff goal is treated as a pair of implications, so both directions of a logical identity arrive in one anonymous constructor. De Morgan and the absorption law:

```
session
λ> :synth (∀ p q : Prop, ¬(p ∨ q) ↔ ¬p ∧ ¬q)
  it1 fun _ _ => {fun k => {fun x => k (.inl x), fun y => k (.inr y)}, fun {k1, k2} z
    ↪ => match z with | .inl w => k1 w | .inr x1 => k2 x1}
λ> :synth (∀ p q : Prop, p ∧ (p ∨ q) ↔ p)
  it1 fun _ _ => {fun {x, _} => x, fun y => {y, .inl y}}
```

Under propositions-as-types, the proof terms *are* plumbing, and the synthesized spellings make that visible: Iff symmetry is a swap, and $\neg(p \wedge q) \leftrightarrow (p \rightarrow \neg q)$ is nothing but currying:

```
session
λ> :synth (∀ p q : Prop, (p ↔ q) → (q ↔ p))
  it1 fun _ _ {f, g} => {g, f}
λ> :synth (∀ p q : Prop, ¬(p ∧ q) ↔ (p → ¬q))
  it1 fun _ _ => {fun k x y => k {x, y}, fun k1 {z, w} => k1 z w}
```

The textbook curiosities are in range too. No proposition is equivalent to its own negation, and the synthesized proof is the classic self-application trick — the term below applies `k` to itself through `f` twice, the propositional shadow of $\omega\omega$:

```
session
λ> :synth (∀ p : Prop, (p ↔ ¬p) → False)
  it1 fun _ {k, f} => k (f (fun x => k x x)) (f (fun y => k y y))
```

5.2 Programs you already know

Some types have one sensible inhabitant, and asking for it by type is quicker than remembering which library corner it lives in. The `bind` of the reader monad, then the `bind` of the state monad:

session

```

λ> :synth ((S → A) → (A → S → B) → S → B)
  it1 fun f g x => g (f x) x
λ> :synth ((S → A × S) → (A → S → B × S) → S → B × S)
  it1 fun f g x => match f x with | ⟨a, b⟩ => g a b
  it2 fun f g x => match f x with | ⟨a, _⟩ => g a x
  it3 fun f g x => match f x with | ⟨a, b⟩ => (match g a x with | ⟨c, _⟩ => ⟨c, b⟩)

```

The first candidate threads the state correctly. Now look at the second: it is type-correct and runs *g* on the *initial* state, throwing away whatever *f* did to it. That is the classic state-threading bug, and the type admits it just as happily. Types alone cannot tell these apart, which is why all candidates are shown and each is one keystroke from a test run. Continuation-passing style works too; this is the bind of the continuation monad:

session

```

λ> :synth ((A → R) → R) → (A → (B → R) → R) → (B → R) → R
  it1 fun f g h => f (fun x => g x h)

```

With the inductive expansion of §5.5 the same game extends to data. Here is `Option.bind`, with the lazy `.none` candidate ranked first and the real one second. When in doubt, run one:

session

```

λ> :synth (∀ a b : Type, Option a → (a → Option b) → Option b)
  it1 fun _ _ _ => .none
  it2 fun _ _ x f => match x with | .none => (.none) | .some y => f y

```

5.3 Rank-N and impredicative goals

A polymorphic hypothesis is not just cargo: Djex instantiates it at types the goal itself supplies — including, under a guard, at *polymorphic* ones. Here the first candidate applies the identity hypothesis to the whole goal $Q \rightarrow Q$, an impredicative instantiation:

session

```

λ> :synth ((∀ p : Prop, p → p) → Q → Q)
  it1 fun f => f _
  it2 fun _ x => x
  it3 fun f x => f _ x

```

And a Church-encoded pair converts into a real conjunction — the quantified hypothesis is instantiated once at *p* and once at *q*, fed the matching projection each time:

session

```

λ> :synth (∀ p q : Prop, (∀ r : Prop, (p → q → r) → r) → p ∧ q)
  it1 fun _ _ f => (f _ (fun x _ => x), f _ (fun _ y => y))

```

(Trailing candidates are elided.) Explicit $\forall$ binders — leading, nested, trailing, or interleaved — are woven into the candidate’s lambda automatically, and uses of quantified hypotheses

get placeholder type arguments wherever Lean needs them (`f _ x`), so bounded rank- N candidates verify. Full impredicative inhabitation is undecidable; beyond the guard the answer is “no term found within bounds” and nothing stronger.

Context-free instantiation chains extend to four leading binders. Leant inserts the inferred type arguments which Djinn leaves implicit, then asks Lean to verify every rendered candidate against the original type:

session

```
λ> :synth (∀ A B C D R : Type, (∀ a b c d : Type, a → b → c → d → R) → A → B → C → D →
  ↳ R)
  it1 fun _ _ _ _ f => f _ _ _ _
  ...
  it5 fun _ _ _ _ f x y z w => f _ _ _ _ x y z w
```

Vacuous local providers can retain the same query-supplied choice even when their result is an opaque Lean type. Here `Demo.Token` becomes a fixed ambient query rigid, not an unresolved inference variable:

session

```
λ> axiom Demo.Token : Type
λ> :synth ((∀ x : Type, x → x) → ({a : Type 1} → Demo.Token) → Demo.Token)
  it1 fun _ x => @x (∀ (a0_0 : Type _), a0_0 → a0_0)
λ> :set synth-engine exference
synth engine: exference
λ> :synth ((∀ x : Type, x → x) → ({a : Type 1} → Demo.Token) → Demo.Token)
  it1 fun f x => f _ (@x (∀ (a0_0 : Type _), a0_0 → a0_0))
```

This visible specialization is available only for a fully vacuous, context-free leading provider chain, and it selects only a closed proper type already present in the checked query. Fixed ambient rigids may occur in the provider result; a free flexible variable rejects the route. A local value has no global binder name, so Leant renders a positional `@` application. For a quantified argument it tries binder domains `_`, `Type _`, and `Prop`, in that order, and backend verification keeps the first spelling which elaborates. Named globals retain source-directed applications. Each domain lane retains at most twelve site-and-style spellings, for a hard 36-variant bound only when the local quantified argument is domain-sensitive; ordinary duplicate spellings collapse to the historical smaller group.

Leant also retains applications whose arguments are proper types. A bound first-order constructor remains a higher-kinded variable in Djex; an opaque Lean family receives one rigid nominal head across its occurrences. Both engines can therefore see a query-supplied polytype without learning any hidden implementation:

session

```
λ> axiom Wrap : Type 1 → Type
λ> :synth ((∀ a : Type 1, Wrap a) → Wrap (∀ b : Type, b → b))
  it1 fun x => x _
λ> :set synth-engine exference
synth engine: exference
```

```
λ> :synth (∀ (F : Type 1 → Type), (∀ a : Type 1, F a) → F (∀ b : Type, b → b))
it1 fun _ x => x _
```

The serializer admits only first-order kind arrows between universes and only application arguments whose own type is a universe. Term-indexed families such as P_3 remain opaque. A retained nominal application can support a positive, Lean-verified candidate but always poisons a negative verdict because its constant hides structure. Candidate types come only from proven proper-type positions below arrows and tuples in the sequent, including quantified subtrees already present there: this is guarded impredicativity, not general second-order search. A visible provider chain is capped at four binders and 32 argument combinations. The separate explicit-instance-head route remains monotype-only; open or context-bearing specified arguments fail closed, and chains with five or more leading binders remain a deliberate incomplete boundary. The implementation contract and Lean 4.31 transcript are recorded in the scoped local-provider report.

5.4 Impossibility, proved

When Djinn’s complete search exhausts a fully translated goal, failure is a theorem:

session

```
λ> :synth (∀ a b : Type, a → b)
provably uninhabited – no closed term of this polymorphic type exists
```

No Lean tactic gives you this answer (a failing `exact?` proves nothing). Note the careful wording: the verdict is about *closed terms of the polymorphic type*. It does not say any particular instance $a \rightarrow b$ is empty (choose $a = b$ and it isn’t), and for propositions it is about *constructive* provability, a distinction §5.7 exploits. When the translation had to hide structure behind an opaque atom, the verdict backs off to match:

session

```
λ> :synth (∀ a : Type, Sigma (fun _ : Nat => a))
no term found within bounds (opaque atoms `Nat` hide structure the engine cannot
↳ analyze – not a refutation)
```

5.5 Inductive types

A non-recursive, non-indexed inductive or structure applied to all its parameters is expanded into a generalized sum of products: constructors become introduction rules, case analysis the elimination rule. This covers the standard library’s little workhorses —

session

```
λ> :synth Ordering
it1 .lt
it2 .eq
it3 .gt
```

```

λ> :synth (∀ a : Type, Option (Option a) → Option a)
  it1 fun _ _ => .none
  it2 fun _ x => match x with | .none => (.none) | .some y => y
λ> :synth (∀ e a b : Type, Except e a → (a → b) → Except e b)
  it1 fun _ _ _ x f => match x with | .error y => .error y | .ok z => .ok (f z)

```

— `Option.join` and `Except.map` synthesized rather than remembered. It extends to Type-valued classes-as-data like `Decidable`, where the eliminations write themselves, up to and including the instance combinators a library author would otherwise write by hand:

session

```

λ> :synth (∀ p : Prop, Decidable p → Decidable (¬ p))
  it1 fun _ x => match x with | .isFalse k => .isTrue k | .isTrue y => .isFalse (fun
    ↪ k1 => k1 y)
λ> :synth (∀ p q : Prop, Decidable p → Decidable q → Decidable (p ∧ q))
  it1 fun _ _ x y => match x with | .isFalse k => .isFalse (fun {z, _} => k z) |
    ↪ .isTrue w => (match y with | .isFalse k1 => .isFalse (fun {_, x1} => k1 x1) |
    ↪ .isTrue x2 => .isTrue {w, x2})
λ> :synth (∀ p q : Prop, Decidable p → Decidable q → Decidable (p → q))
  it1 fun _ _ x y => match x with | .isFalse k => .isTrue (fun z => absurd z k) |
    ↪ .isTrue w => (match y with | .isFalse k1 => .isFalse (fun f => k1 (f w)) |
    ↪ .isTrue x1 => .isTrue (fun _ => x1))

```

The last two are decidability of conjunction and of implication, assembled by case analysis on both instance arguments: four branches each, every branch carrying its own little proof.

Session-declared types participate the moment you declare them, and refutations over expanded inductives remain sound, because the engine saw the complete constructor list:

session

```

λ> structure Pair (A B : Type) where
...>   fst : A
...>   snd : B
...>
λ> :synth (∀ a b : Type, a → b → Pair a b)
  it1 fun _ _ x y => {x, y}
λ> :synth (∀ a b : Type, Pair a b → Empty)
provably uninhabited – no closed term of this polymorphic type exists

```

Recursive types cannot be expanded; elimination is where undecidability lives. Their *constructors* are still sound introduction rules, though, so building is possible even where analysis is not:

session

```

λ> :synth Nat
  it1 Nat.zero
  it2 Nat.succ Nat.zero
λ> :synth (∀ a : Type, a → List a)
  it1 fun _ _ => List.nil
  it2 fun _ x => List.cons x List.nil

```

5.6 Recursion from the library

`:synth` will never invent a `Nat.rec`-based program, but for the everyday recursive types it does not have to: the library already wrote the recursion. A goal that mentions `List` or `Nat` brings a rated inventory of library functions with it — `List.map`, `List.foldr`, `List.append`, `List.flatten`, `List.length`, `List.replicate`, `Nat.add`, ... — instantiated at the goal's own types and offered to the engine as extra premises. The enumeration prefers proofs that use the goal's own arguments, so asking for a function *by its type* puts the library answer first:

session

```
λ> :synth ((a → b) → List a → List b)
  it1 fun f x => List.map f x
  it2 fun f x => List.reverse (List.map f x)
λ> :synth (List (List a) → List a)
  it1 fun x => List.flatten x
  it2 fun x => List.reverse (List.flatten x)
λ> :synth (Nat → a → List a)
  it1 fun x y => List.replicate x y
  it2 fun x y => List.reverse (List.replicate x y)
λ> :synth (List a → List b → List (a × b))
  it1 fun x y => List.zip x y
  it2 fun _ _ => List.nil
λ> :synth ((a → b → c) → List a → List b → List c)
  it1 fun f x y => List.zipWith f x y
  it2 fun _ _ _ => List.nil
```

(Trailing candidates and the truncation notes are elided here.) The inventory is a ratings list in Djex's `*.ratings` format — lower is better, 100 or more disables — and a project file `leant.ratings` (lines of `Name Rating, # comments`) merges over the defaults at startup, so re-ranking, disabling, or growing the inventory is editing a list, not writing code.

The library search runs beside the constructor search of the previous section, in a mode where the recursive occurrences are sealed atoms: without `nil/cons` in play, every candidate must route through the goal's own arguments and the offered functions, which is what keeps `List.map` ahead of the endless supply of closed terms that ignore the input. Both searches' survivors are shown together, library candidates first; `:set synth-library off` restores the constructors-only behavior. As always, the premises are only *offers*: every candidate is re-elaborated by the backend, so a useless premise costs search time, never a wrong answer, and negative verdicts never come from the library run at all.

5.7 Classical candidates

Constructive logic goes further than newcomers expect, and `:synth` stays inside it whenever it can. Triple negation collapses, and contraposition holds in one direction, with no axioms in sight:

session

```
λ> :synth (∀ p : Prop, ¬¬¬p → ¬p)
  it1 fun _ k x => k (fun k1 => k1 x)
```

```
λ> :synth (∀ p q : Prop, (p → q) → ¬q → ¬p)
it1 fun _ _ f k x => k (f x)
```

Peirce's law is different: it has no constructive inhabitant, and `:synth` can prove that. But rather than stopping at the refutation, it then asks the classical question, and answers with a term whose spelling shows what was used:

session

```
λ> :synth (∀ p q : Prop, ((p → q) → p) → p)
it1 fun _ _ f => match Classical.em _ with | .inl x => x | .inr k => f (fun y =>
  ↳ absurd y k)
λ> :synth (∀ p : Prop, ¬¬p → p)
it1 fun _ k => match Classical.em _ with | .inl x => x | .inr k1 => absurd k1 k
```

These are the proofs a human would write: one case split on excluded middle, one `absurd` where the contradiction lands. The hard direction of De Morgan needs excluded middle twice, once per disjunct, and the candidate reads as exactly that case analysis:

session

```
λ> :synth (∀ p q : Prop, ¬(¬p ∧ ¬q) → p ∨ q)
it1 fun _ _ k => match Classical.em _ with | .inl x => .inl x | .inr k1 => (match
  ↳ Classical.em _ with | .inl y => .inr y | .inr k2 => absurd {k1, k2} k)
```

Less famous gaps between the two logics come out the same way. Weak excluded middle, and the Gödel–Dummett formula:

session

```
λ> :synth (∀ p : Prop, ¬p ∨ ¬¬p)
it1 fun _ => match Classical.em _ with | .inl x => .inr (fun k => k x) | .inr k1 =>
  ↳ .inl k1
λ> :synth (∀ p q : Prop, (p → q) ∨ (q → p))
it1 fun _ _ => match Classical.em _ with | .inl _ => (match Classical.em _ with |
  ↳ .inl x => .inl (fun _ => x) | .inr k => .inr (fun y => absurd y k)) | .inr k1 =>
  ↳ .inl (fun z => absurd z k1)
```

Two searches stand behind this. First, one excluded-middle assumption per atomic subformula is handed to the intuitionistic engine — complete for the propositional fragment, since intuitionistic logic plus atom instances of $p \vee \neg p$ is classical propositional logic. The findings are rendered as `Classical.em` case splits. If that misses, the goal's double-negation translation runs instead (complete by Glivenko's theorem) and the candidate is wrapped in `Classical.byContradiction`. Both kinds are verified like any other candidate. The whole behavior sits behind a switch:

session

```
λ> :set synth-classical off
synth classical: off
λ> :synth (∀ p : Prop, p ∨ ¬p)
```


provably uninhabited – no closed term of this polymorphic type exists
(constructively – a classical proof may still exist; this is not a disproof of the
↪ proposition)

5.8 Dependent formulas as cargo

Leant's translation makes no attempt to analyze dependent types. It does not refuse them either: a dependent subformula becomes an opaque *atom*, compared up to α -equivalence, which the engine can carry from one side of a goal to the other. Transport works; analysis does not:

session

```
λ> opaque P : Nat → Prop
λ> opaque Q : Prop
λ> :synth ((∀ n : Nat, P n) ∧ Q → Q ∧ (∀ n : Nat, P n))
  it1 fun (x, y) => {y, x}
```

The engine never looked inside $\forall n, Pn$; it swapped a sealed box. A goal that would require opening the box (an induction, a rewrite, a case split on an index) is refused with a reason; that work belongs to :prove.

5.9 Synthesis inside a proof

Bare :synth in prove mode targets the current goal *with its hypotheses as premises*. Candidates are offered as it1, it2, ... exactly as in the session, except that here itN splices the candidate applied to your hypotheses, so exact it1 closes the goal. The hypotheses can carry real weight; below, the case analysis on the Decidable instance comes from hd:

session

```
λ> :prove ∀ p q : Prop, Decidable p → (p → q) → (¬p → q) → q
entering prove mode – type tactics; :help for commands
⊢ ∀ (p q : Prop) (a : Decidable p), (p → q) → (¬p → q) → q
suggestion: exact fun p q a a_1 a_2 => dite p a_1 a_2 (closes the goal)
⊢> intro p q hd hpq hnq
  p q : Prop
  hd : Decidable p
  hpq : p → q
  hnq : ¬p → q
⊢ q
suggestion: exact dite p hpq hnq (closes the goal)
⊢> :synth
(synthesizing with hypotheses p q hd hpq hnq as premises)
  it1 fun _ _ x f g => match x with | .isFalse k => g k | .isTrue y => f y
⊢> exact it1
All goals accomplished
finish with :qed [NAME], inspect with :script
⊢> :qed byCases
saved: theorem byCases : ∀ p q : Prop, Decidable p → (p → q) → (¬p → q) → q
```

Note the contrast with the tactic suggestion above it: the suggestion *found* the library's dte, while `:synth` *composed* the case split from the goal's own material. Both are verified; they simply answer different questions.

It also slots into an ordinary multi-goal proof. Split a conjunction with `constructor` and let `:synth` deal with each half in turn. Notice the prompt tracking the number of open goals, and the projections from `h` showing up without being asked for:

```

session
λ> :prove ∀ p q r : Prop, (p → q) ∧ (q → r) → p → r ∧ (p → q)
entering prove mode – type tactics; :help for commands
⊢ ∀ (p q r : Prop), (p → q) ∧ (q → r) → p → r ∧ (p → q)
suggestion: intro p q r h h1 <;> simp_all (closes the goal)
⊢> intro p q r h hp
p q r : Prop
h : (p → q) ∧ (q → r)
hp : p
⊢ r ∧ (p → q)
suggestion: simp_all only [forall_const, imp_self, and_self] (closes the goal)
⊢> constructor
– goal 1 of 2 –
case left
p q r : Prop
h : (p → q) ∧ (q → r)
hp : p
⊢ r
– goal 2 of 2 –
case right
p q r : Prop
h : (p → q) ∧ (q → r)
hp : p
⊢ p → q
suggestion: simp_all only [forall_const] (closes the goal)
⊢> :synth
(synthesizing with hypotheses p q r h hp as premises)
  it1 fun _ _ _ (f, g) x => g (f x)
⊢> exact it1
case right
p q r : Prop
h : (p → q) ∧ (q → r)
hp : p
⊢ p → q
suggestion: simp_all only [forall_const, imp_self] (closes the goal)
⊢> :synth
(synthesizing with hypotheses p q r h hp as premises)
  it1 fun _ _ _ (f, _) x _ => f x
⊢> exact it1
All goals accomplished
finish with :qed [NAME], inspect with :script
⊢> :qed chain
saved: theorem chain : ∀ p q r : Prop, (p → q) ∧ (q → r) → p → r ∧ (p → q)

```

Unlike `exact?`, which finds an *existing* lemma, this composes a new term from the goal's own material: a constructive complement to the finisher tactics, needing no premise database

and no imports. The division of labor that emerges is pleasant: the genuinely dependent moves (quantifiers, induction, rewriting) stay yours, and the propositional bookkeeping between them does not.

The classical fallback of §5.7 follows `:synth` into prove mode. Double-negation elimination has no constructive proof, so the candidate arrives as an excluded-middle case split, and `:qed` turns it into a named theorem — no tactic beyond `exact` was typed:

session

```
λ> :prove ∀ p : Prop, ¬¬p → p
entering prove mode — type tactics; :help for commands
⊢ ∀ (p : Prop), ¬¬p → p
suggestion: exact fun p a => Classical.byContradiction a (closes the goal)
⊢> :synth
  it1 fun _ k => match Classical.em _ with | .inl x => x | .inr k1 => absurd k1 k
⊢> exact it1
All goals accomplished
finish with :qed [NAME], inspect with :script
⊢> :qed not_not_elim
saved: theorem not_not_elim : ∀ p : Prop, ¬¬p → p
```

5.10 Engines and budgets

The default engine is Djinn: complete, terminating, and the only source of refutations. `:set synth-engine exference` switches to the ranked heuristic search, and both runs the two together — negative verdicts still come only from Djinn. Standalone lanes send at most 12 fresh candidate groups to Lean. A combined lane gets 24 and retains both standalone frontiers: writing D and E for fresh Djinn and Exference groups, its order is D1–D4, E1–E12, D5–D12, then alternating tails. Exference runs under explicit budgets (`:set synth-steps`, default 4096), and truncation is always reported:

session

```
λ> :set synth-engine both
synth engine: both
λ> :synth ((A → B → C) → (A → B) → A → C)
  it1 fun f g x => f x (g x)
note: exference: search truncated: step limit reached, queue limit pruned 31990
```

Every engine mode first gives a structurally accepted goal a provider-free lane and asks Lean to verify its candidates. After a nonterminal miss, Lean discovers at most 80 live term providers under the namespaces used by the target (plus `exact` session declarations). Type constructors and type families, whose fully peeled result is a sort, are excluded before search. Exact-result session and public declarations lead the ranking; conventional implementation-worker names remain eligible but follow public fallbacks, while session declarations bypass that naming heuristic. Exference searches the ranked inventory directly; Djinn tries discovery-order prefixes of 1, 4, and 16 providers before the full bounded inventory, and may specialize a loaded scheme at a closed monotype or guarded rank-N polytype. The serializer derives a canonical query from the sorted, deduplicated namespace roots and final result head, rather

than from the goal’s raw spelling. Successful inventories — including an empty one — are kept in a generation-aware 12-entry LRU. Imports, ordinary declarations, and session-restoring operations invalidate that generation; generated `it1`, `it2`, ... bindings cannot be providers and preserve it.

Exact rendered spellings keep their first scheduled occurrence. Before a later provider lane is forced or capped, spellings already rejected by Lean are removed from each engine’s source stream and newly empty groups are dropped. Rediscovered failures therefore spend none of that lane’s 12- or 24-group fresh frontier.

The pure searches answer in microseconds to milliseconds; the cost center is backend verification, a few hundred milliseconds per candidate. A wall-clock guard (default 20 s, `LEANT_SYNTH_TIMEOUT`) covers the quantified goals whose bounded instantiation can widen the space; hitting it is reported as “no answer,” never as a verdict.

5.11 What `:synth` will not do

Some plain statements of present limits. It will not synthesize recursion or induction. Complete compatible recursive families contribute bounded positive construction to Djinn and one constructor layer of elimination to Exference; recursive fields are never opened again. It will not analyze dependent structure; atoms are transported, never opened. Opaque and bound-variable proper-type applications retain their ordered arguments, and qualifying inductives share one exact parametric family across distinct `Option/List` occurrences. Term-indexed, dependent, ambiguous, or incompatible families keep their conservative fallback. Quantifier handling beyond the propositional fragment follows Djex’s bounded rank-N and guarded impredicativity rules, which are sound and terminating over an undecidable space and widen as Djex improves. The current plan family is exhaustive through seven independent quantified sites; an eight-site four-open/four-opaque goal is the next deliberate occurrence-plan gap, while hypothesis instantiation chains with five or more binders remain opaque. Such higher-rank goals answer “no term found within bounds” and nothing stronger. And although Djex can resolve type-class evidence itself, Leant leaves instance arguments to Lean’s elaborator during verification, which sits nearer the instance tables anyway. A non-dependent instance-implicit goal binder is retained as a render-only slot: it contributes no dictionary premise to either engine, but Leant inserts its wildcard before later lambda arguments and withholds complete negative evidence across the erased boundary. Constrained rank-N hypotheses therefore keep their instance applications implicit without shifting the candidate’s term binders. Provider projection does not serialize Lean’s instance registry into Djex, so a polytype visible only inside an instance head is not a candidate: it must already occur in the checked query. Extending that boundary would require bounded, checked caller-supplied evidence rather than an approximation through Haskell classes. The fragment is bounded on purpose: inside it, answers are fast and verified. Refutations are final for the complete provider-free structural calculus; the bounded, best-effort live-provider inventory is not an exhaustive theorem about every declaration or axiom in the Lean environment.

6 How it works, briefly

A `:synth` query passes through five checked stages. A Lean metaprogram (compiled once into a cached side environment) elaborates the goal and serializes it into the engines' fragment: structural connectives are decomposed; qualifying inductives expand into constructor lists; proper-type applications retain rigid or bound heads and ordered arguments; everything else becomes an α -normalized opaque atom. The fragment translator either accepts the result or refuses with a reason. The cached synthesis environment remembers the session history it has replayed: an unchanged history is reused, an append replays only its new suffix, and undo or another non-prefix change rebuilds from the cached import-and-serializer base. Generated result bindings participate in this replay so later goals can mention them without flushing the provider cache. The selected engine searches; its candidates are rendered back into Lean syntax, with constructor names restored, binders named by role, and anonymous-constructor and leading-dot spellings preferred over fully qualified fallbacks. Finally the backend re-elaborates each candidate against the original goal, and only survivors are shown and bound. The engine boundary is narrow by design: the searches behind it are swappable, and nothing they emit is believed until Lean has type-checked it.

An unpickled environment is an explicit history barrier. Leant copies it to a private process-lifetime artifact so backend restart can restore that base before replaying the commands entered afterward; the reconstructed undo stack therefore still bottoms out at the snapshot. `:reset` and `:load` leave snapshot mode, whereas `:import` requires that explicit transition because imports cannot be added to an opaque Lean environment. A Leant `:pickle` also writes a content-fingerprinted, serializer-ABI-keyed synthesis companion. Foreign or stale sidecars are ignored without rejecting the main upstream snapshot; if that snapshot exposes Lean's metaprogramming API, Leant compiles current synthesis tooling directly over it.

7 Status

Leant is a working prototype under active development. The synthesis design and its phased history are documented in `docs/SYNTHESIS_PROPOSAL.md`; transcript-level golden tests live in `test/`. The project began inside the *ProveIt* repository and was split out with its history intact. Expect sharp edges: the project exists to find out how far a conversational, verification-gated synthesis workflow can be pushed, and it moves at the speed of that question.